WORKED SOLUTION

STEP I 1995, Question 5

Pure • Binomial Series and Integration

STEP I 1995, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s1-q5

The key idea

Differentiating a polynomial term by term kills the awkward $1/r$ factors, and what is left is almost a binomial expansion: the coefficients $(-1)^{r+1}\binom{n}{r}$ are exactly those of $-(1-x)^n$ with its constant term removed. Once f' is known, the fundamental theorem of calculus (with $\mathrm{f}(0)=0$) gives f as an integral, and a substitution $y = 1-t$ turns it into the required form. The integrand there is a geometric series, which integrates at once.

Solution

The derivative

Write

$$\mathrm{f}(x) = \sum_{r=1}^{n} (-1)^{r+1} \binom{n}{r} \frac{x^r}{r}.$$

This is a polynomial, so we may differentiate term by term:

$$\mathrm{f}'(x) = \sum_{r=1}^{n} (-1)^{r+1} \binom{n}{r} x^{r-1}.$$

For $x \neq 0$, multiply by x and use $(-1)^{r+1}x^r = -(-x)^r$:

$$\begin{aligned} x\,\mathrm{f}'(x) &= -\sum_{r=1}^{n} \binom{n}{r} (-x)^r \\ &= -\Big(\sum_{r=0}^{n} \binom{n}{r} (-x)^r - 1\Big) \\ &= 1 - (1-x)^n, \end{aligned}$$

by the binomial theorem. Dividing by x,

$$\mathrm{f}'(x) = \frac{1-(1-x)^n}{x} \qquad (x \neq 0),$$

as required. (At $x = 0$ the series gives $f'(0) = n$, which is also the limit of the right-hand side as $x \to 0$, since $1 - (1-x)^n = nx + O(x^2)$.)

The integral formula

Every term of f contains a factor x, so $f(0) = 0$. By the fundamental theorem of calculus,

$$f(x) = f(0) + \int_0^x f'(t)\,dt = \int_0^x \frac{1-(1-t)^n}{t}\,dt.$$

> **Ask yourself** Is it safe to integrate through $t = 0$, where the formula has t in the denominator?

Yes. The integrand is really the polynomial $f'(t)$; the fraction is just a compact way of writing it for $t \neq 0$, and its value at $t = 0$ is the limit n. So there is no singularity, only a removable gap in the formula.

Now substitute $y = 1 - t$, so $dt = -dy$; when $t = 0$, $y = 1$, and when $t = x$, $y = 1 - x$. Then

$$\begin{aligned} f(x) &= \int_1^{1-x} \frac{1-y^n}{1-y}\,(-dy) \\ &= \int_{1-x}^1 \frac{1-y^n}{1-y}\,dy, \end{aligned}$$

reversing the limits to absorb the minus sign. This is the required result, and it holds for every real x.

The value of $f(1)$

Put $x = 1$, so the lower limit is 0. For $y \neq 1$ the integrand is the sum of a geometric series,

$$\frac{1-y^n}{1-y} = 1 + y + y^2 + \cdots + y^{n-1},$$

and (as before) the value at $y = 1$ is the limit n, so the integrand is this polynomial on the whole of $[0, 1]$. Hence

$$\begin{aligned} f(1) &= \int_0^1 (1 + y + \cdots + y^{n-1})\,dy \\ &= \Big[y + \frac{y^2}{2} + \cdots + \frac{y^n}{n} \Big]_0^1 \\ &= 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}, \end{aligned}$$

as required.

Notice what this says: putting $x = 1$ in the original series gives the striking identity

$$\sum_{r=1}^{n} \frac{(-1)^{r+1}}{r} \binom{n}{r} = 1 + \frac{1}{2} + \cdots + \frac{1}{n},$$

which is far from obvious directly.

Checks

- $n = 1$: $\mathrm{f}(x) = x$, and $\big(1 - (1 - x)\big)/x = 1 = \mathrm{f}'(x)$; also $\mathrm{f}(1) = 1$.
- $n = 4$: $\mathrm{f}(1) = 4 - \frac{6}{2} + \frac{4}{3} - \frac{1}{4} = \frac{25}{12} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}$.
- We checked the derivative formula exactly at 300 random rational points, compared the integral (by Simpson's rule) with the series for random $x \in [-1.5, 1.5]$ and $n \leq 15$ (relative agreement better than 10^{-11}), and confirmed the identity for $\mathrm{f}(1)$ exactly for every $n \leq 40$.

Common slips

- Forgetting the $r = 0$ term when matching with the binomial expansion of $(1 - x)^n$: the sum from $r = 1$ is $(1 - x)^n - 1$, not $(1 - x)^n$.
- Writing $\mathrm{f}(x) = \int_0^x \mathrm{f}'(t)\,\mathrm{d}t$ without noting that $\mathrm{f}(0) = 0$.
- Getting the limits the wrong way round after the substitution $y = 1 - t$: the sign of $\mathrm{d}t = -\mathrm{d}y$ is what flips them back.

Beyond STEP

The numbers $H_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n}$ are the harmonic numbers; they grow like $\ln n$, and $H_n - \ln n$ tends to Euler's constant $\gamma \approx 0.5772$. The trick used here, differentiating to remove awkward denominators and then integrating back, is the same one that gives the power series for $\ln(1 + x)$ and $\tan^{-1} x$, and its justification for infinite series (term-by-term differentiation of power series) is a core topic of first-year analysis. Abbott's *Understanding Analysis* treats this carefully.

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